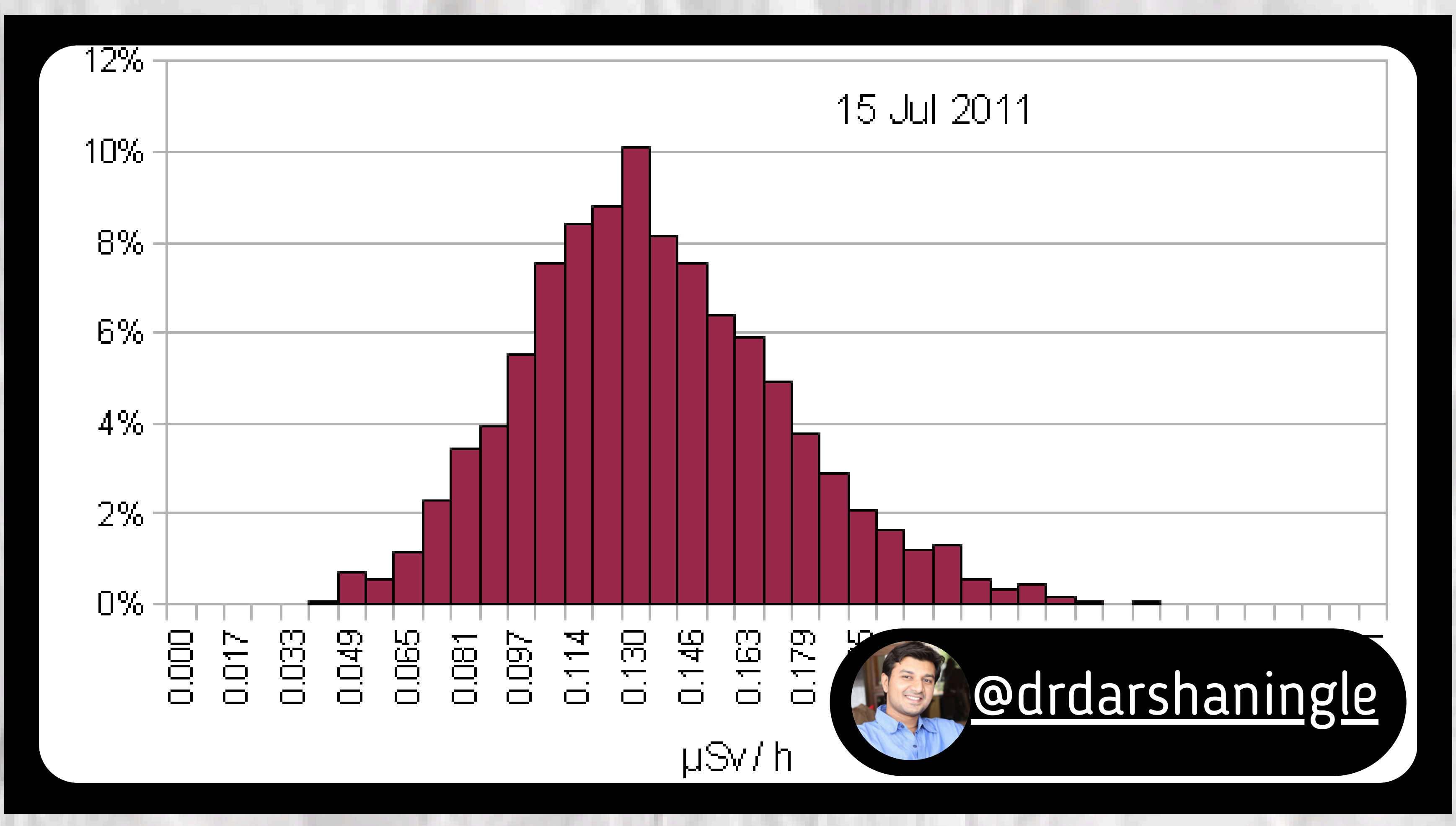
HISTOGRAM



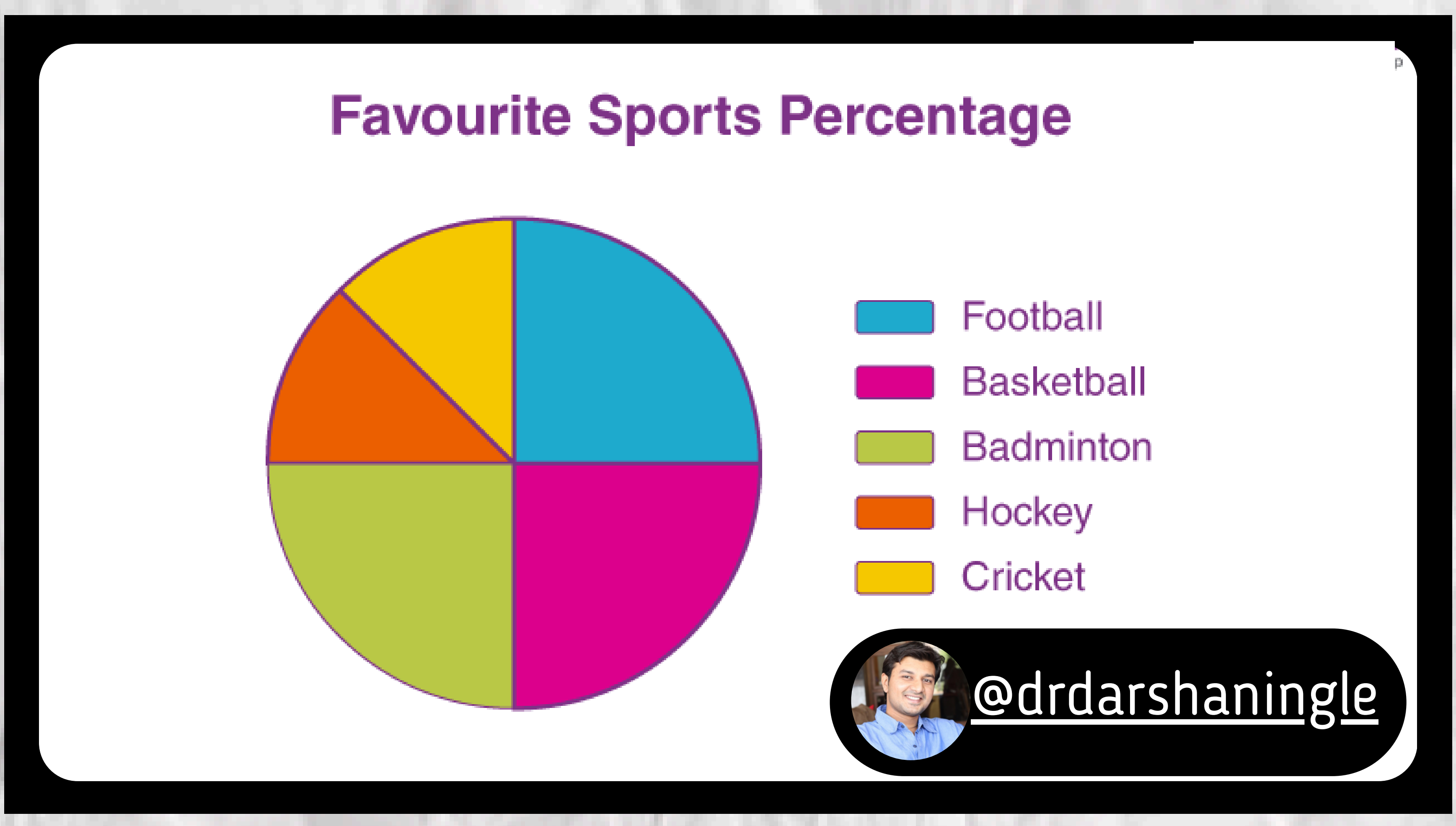
Data Type : Univariate

USE WHEN: VISUALIZING THE DISTRIBUTION OF NUMERICAL DATA.

DON'T USE WHEN NOT FOR COMPARING CATEGORIES OR TRENDS.

EXAMPLE : DISTRIBUTION OF EXAM SCORES, CUSTOMER AGE DEMOGRAPHICS.

PIE CHART



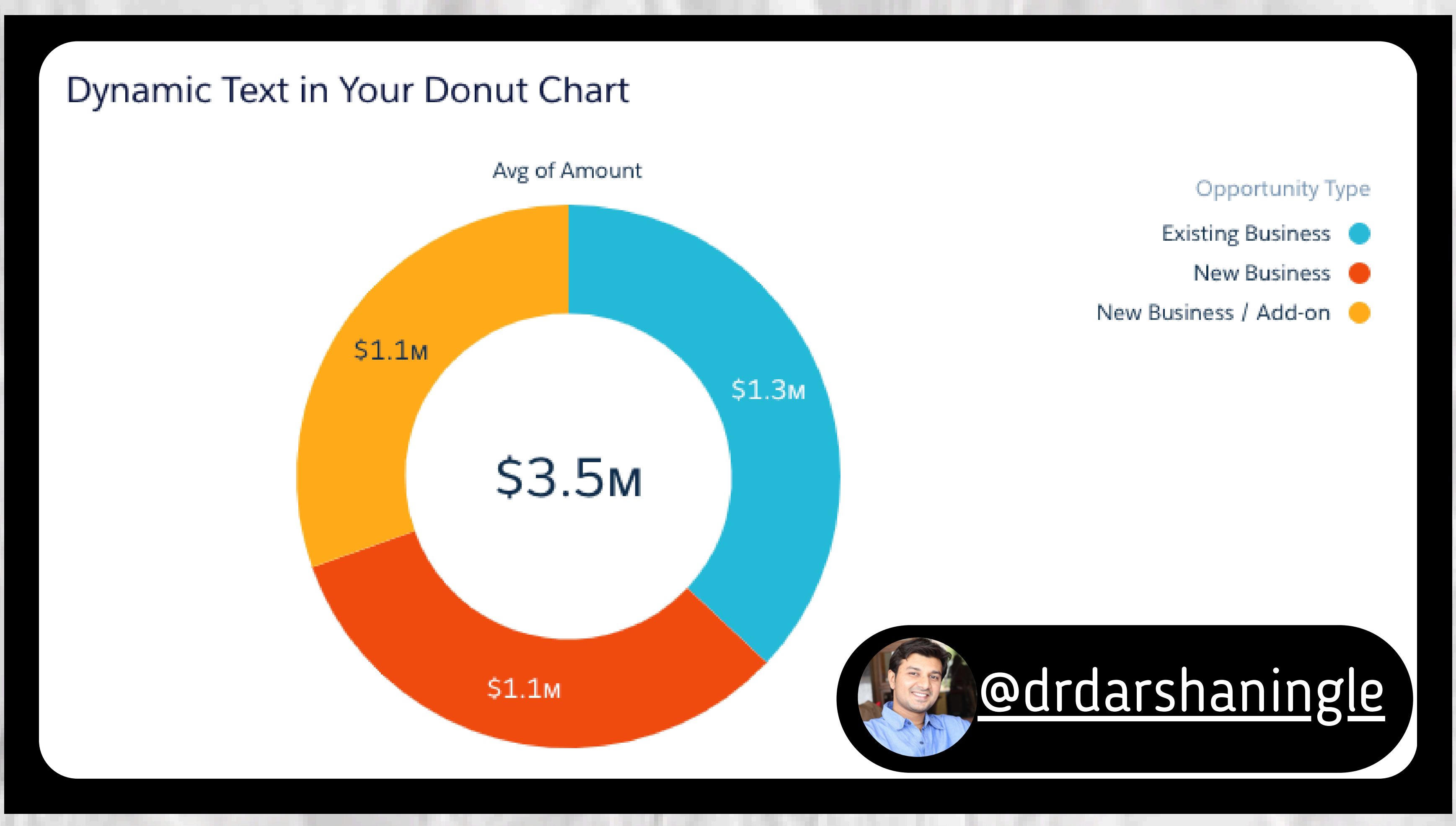
Data Type : Univariate

USE WHEN: SHOWING THE COMPOSITION OF A WHOLE, LIMITED DATA POINTS.

DON'T USE: WHEN MORE THAN 5-7 DATA POINTS, COMPLEX COMPARISONS.

EXAMPLE : BUDGET BREAKDOWN BY EXPENSE CATEGORY, MARKET SHARE OF DIFFERENT BRANDS

DONUT CHART



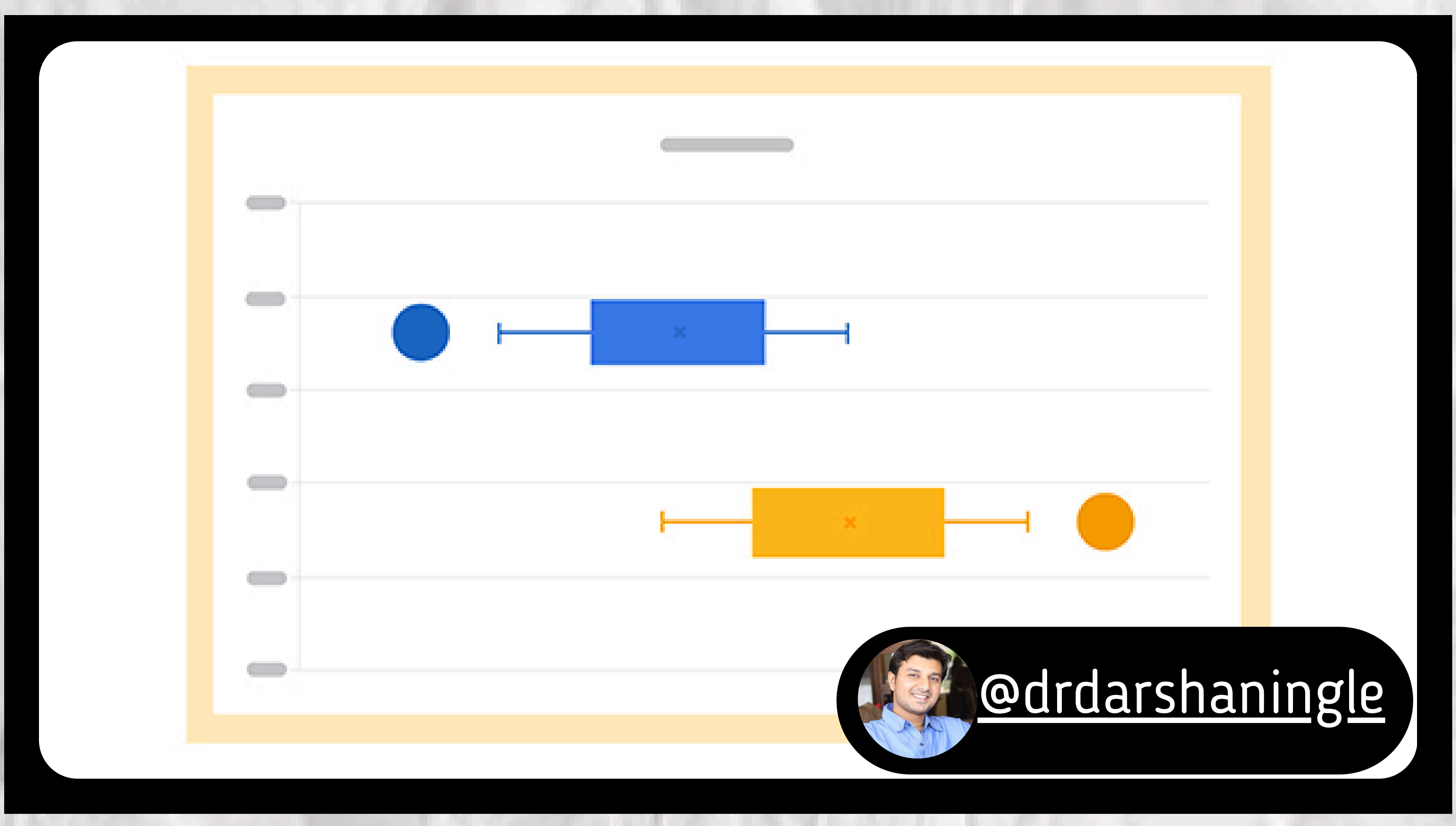
Data Type : Univariate

USE WHEN: LIMITED DATA POINTS WITH EMPHASIS ON A SINGLE ELEMENT.

DON'T USE: WHEN MORE THAN 5-7 DATA POINTS, COMPLEX COMPARISONS.

EXAMPLE : CUSTOMER SATISFACTION RATINGS (VERY SATISFIED HIGHLIGHTED).

BOX PLOT



Data Type : Univariate

USE WHEN: DISPLAYING THE DISTRIBUTION OF DATA WITH QUARTILES AND OUTLIERS.

DON'T USE WHEN SMALL DATASETS, COMPLEX COMPARISONS.

EXAMPLE : EMPLOYEE SALARIES BY DEPARTMENT (SHOWING OUTLIERS).

WORD CLOUD

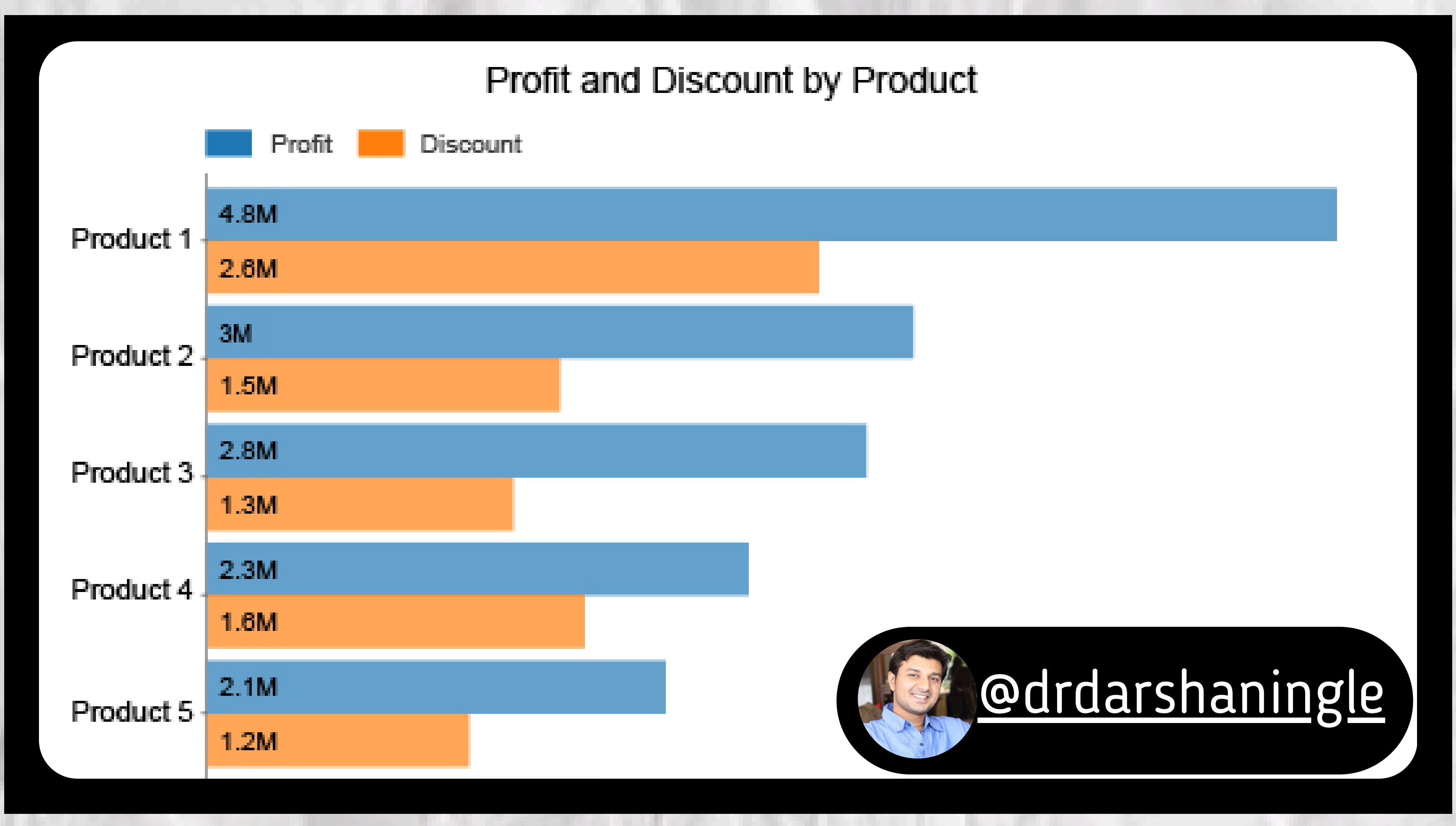
Data Type : Univariate

USE WHEN: EMPHASIZING WORD FREQUENCY AND PROMINENCE IN TEXTUAL DATA.

DON'T USE WHEN: COMPLEX DATA ANALYSIS, PRECISE COMPARISONS.

EXAMPLE : POPULAR HASHTAGS ON SOCIAL MEDIA, KEYWORDS USED IN CUSTOMER REVIEWS.

BAR CHART



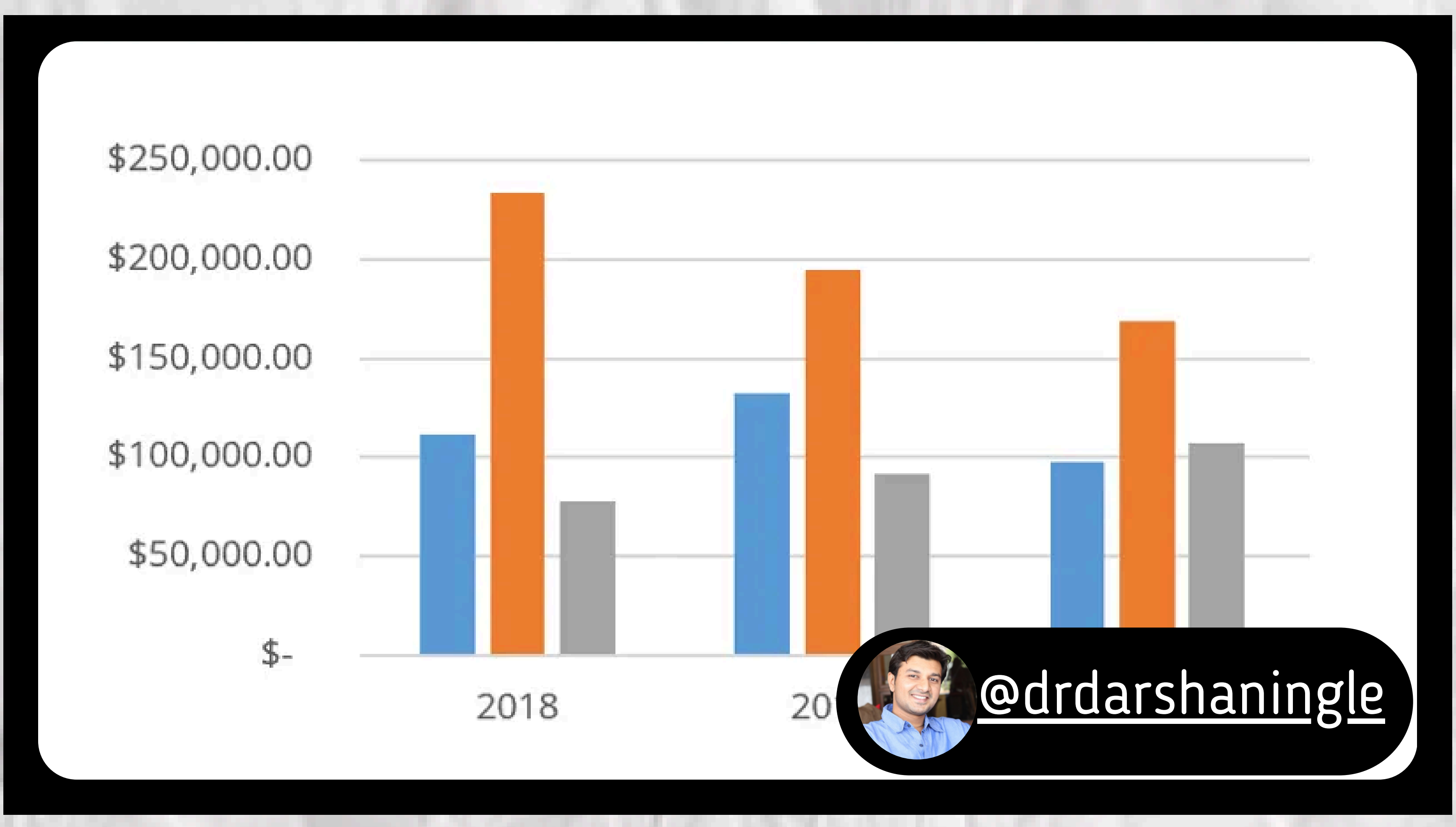
Data Type : Bivariate

USE WHEN: COMPARING CATEGORIES, SHOWING CHANGES OVER TIME.

DON'T USE WHEN: MANY CATEGORIES, COMPLEX COMPARISONS.

EXAMPLE : SALES FIGURES BY PRODUCT CATEGORY, WEBSITE TRAFFIC BY MONTH.

COLUMN CHART



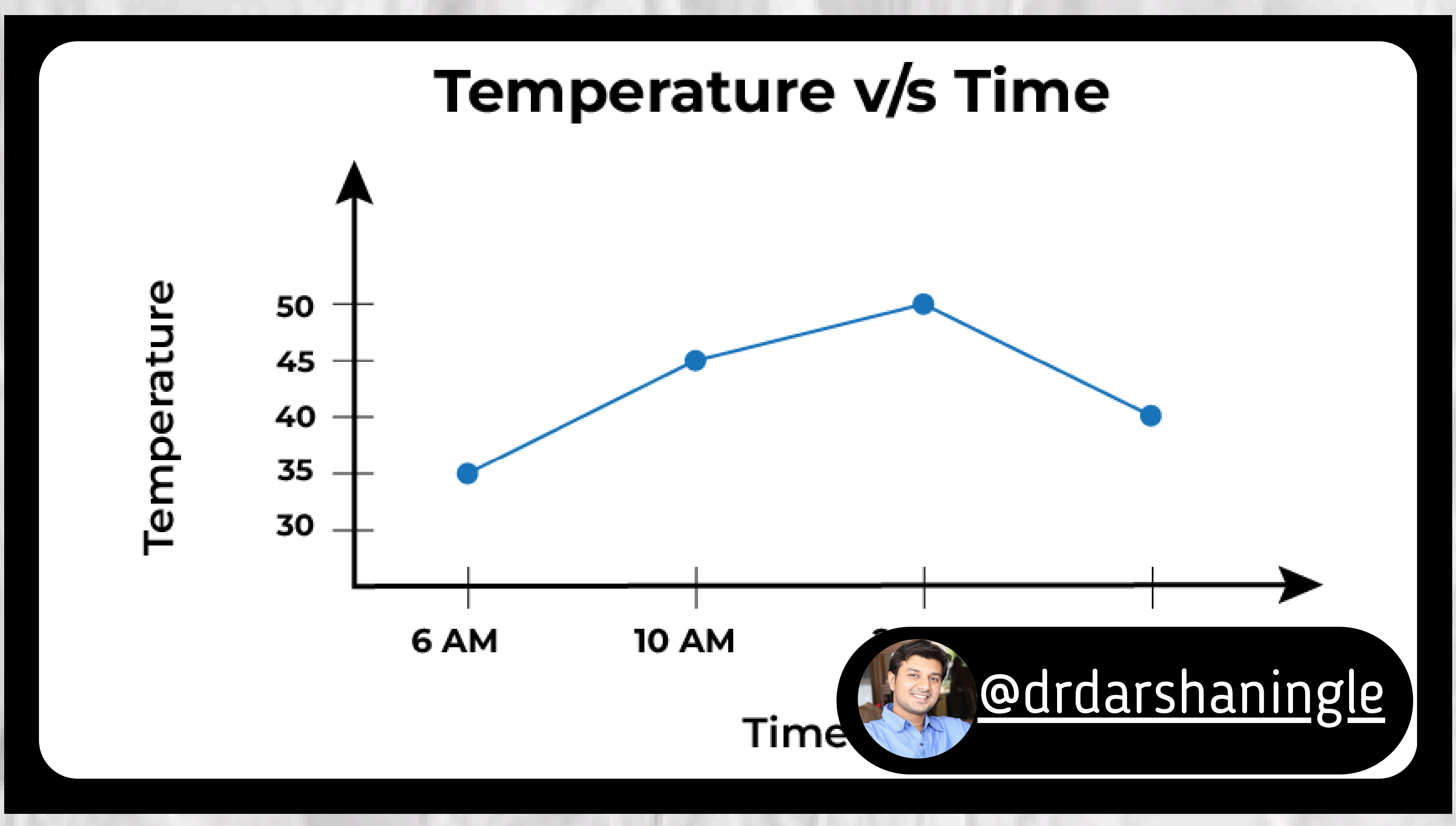
Data Type : Bivariate

USE WHEN: SIMILAR TO BAR CHARTS, BUT BETTER FOR LONG CATEGORY LABELS.

DON'T USE WHEN: MANY CATEGORIES, COMPLEX COMPARISONS.

EXAMPLE : CUSTOMER SATISFACTION BY DEPARTMENT, STOCK PRICES OVER TIME.

LINE CHART



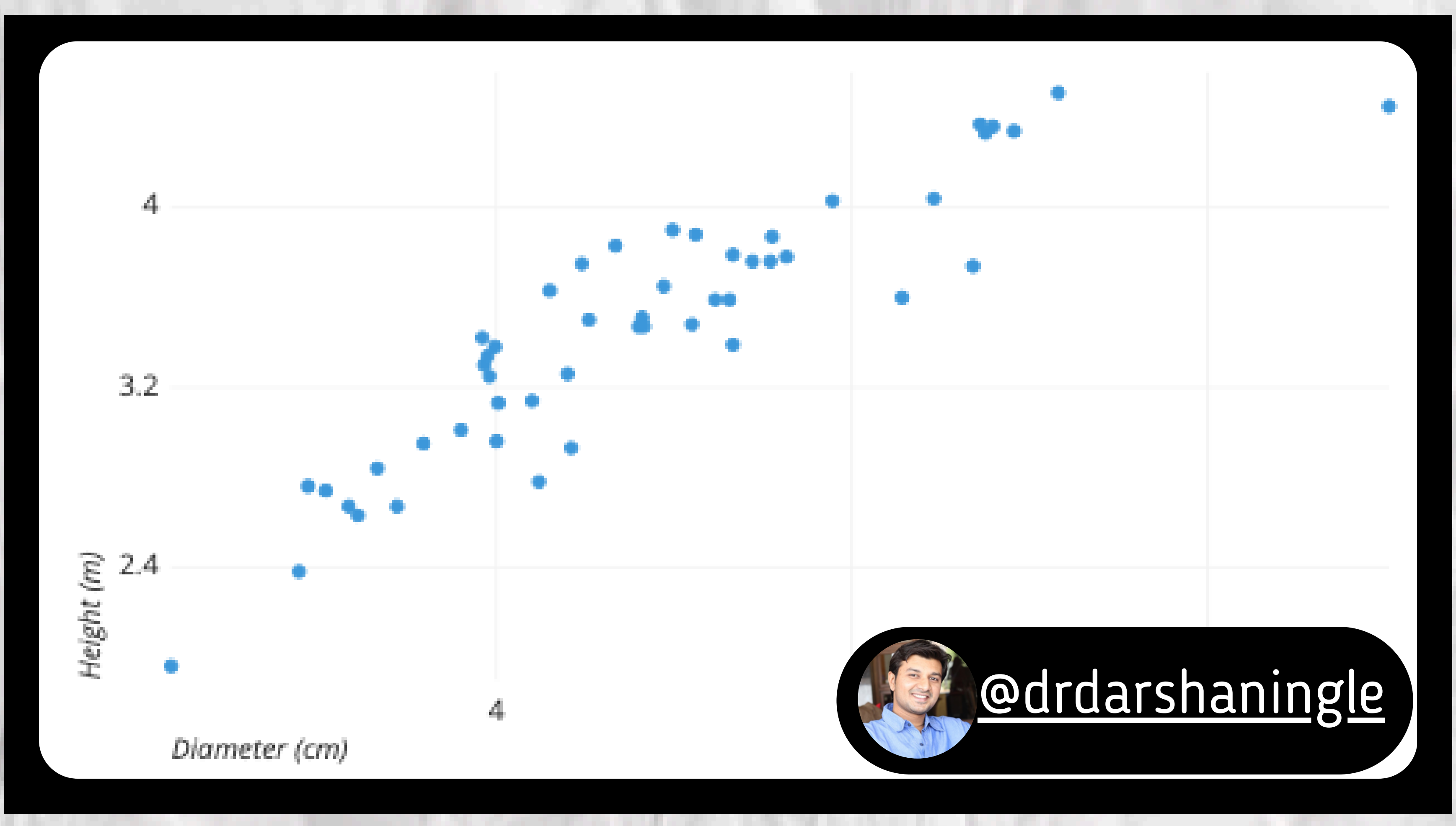
Data Type : Bivariate

USE WHEN: SHOWING TRENDS OVER TIME, CONNECTING DATA POINTS.

DON'T USE WHEN: EMPHASIZING SPECIFIC DATA POINTS, COMPARING CATEGORIES.

EXAMPLE : TEMPERATURE CHANGES THROUGHOUT THE DAY, WEBSITE TRAFFIC GROWTH OVER A YEAR.

SCATTER PLOT



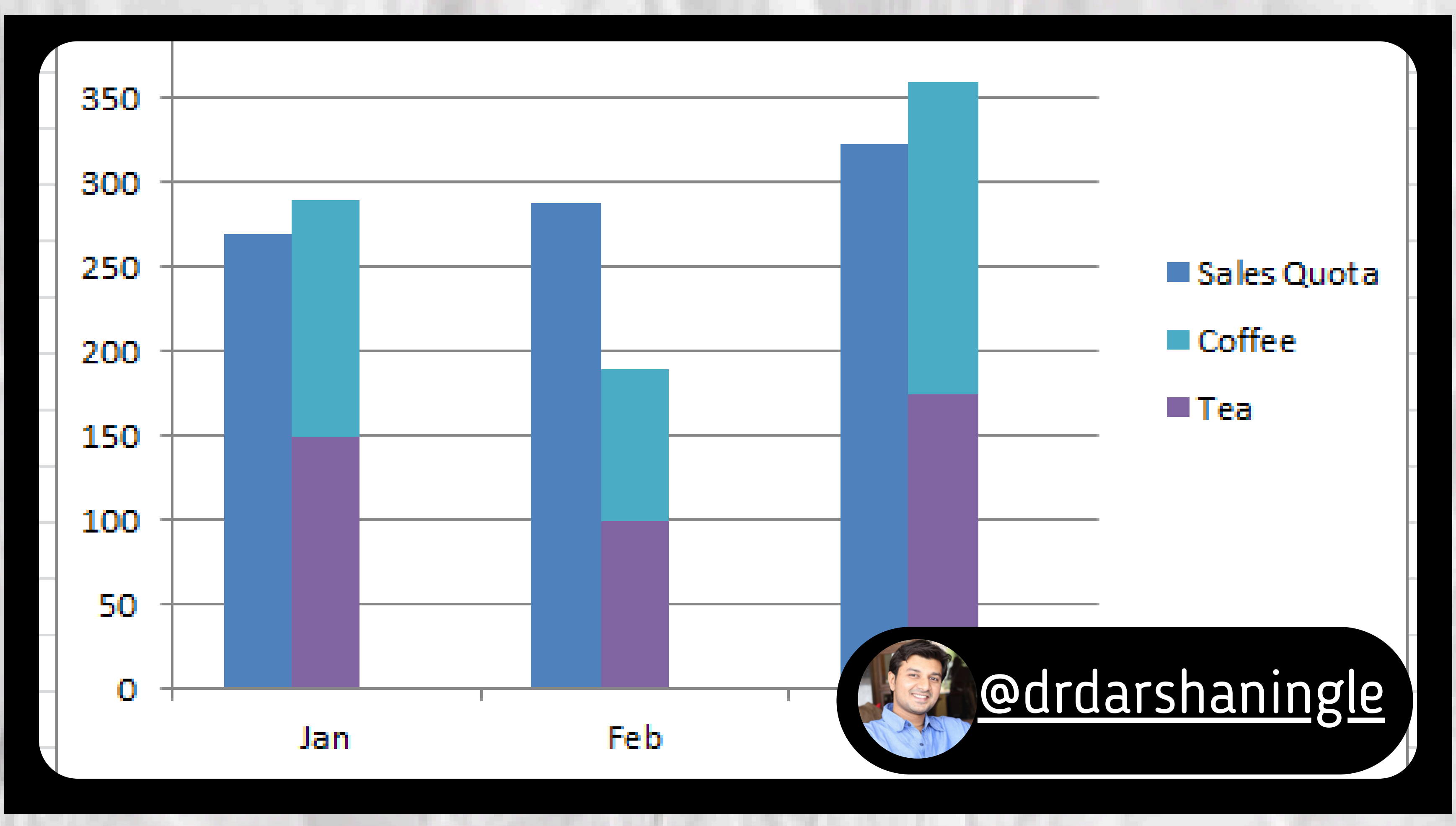
Data Type : Bivariate

USE WHEN: IDENTIFYING RELATIONSHIPS BETWEEN TWO VARIABLES.

DON'T USE WHEN: SHOWING TRENDS OVER TIME, COMPARING CATEGORIES.

EXAMPLE : CUSTOMER AGE VS. PURCHASE AMOUNT, EXAM SCORES VS. STUDY HOURS.

STACKED COLUMN CHART



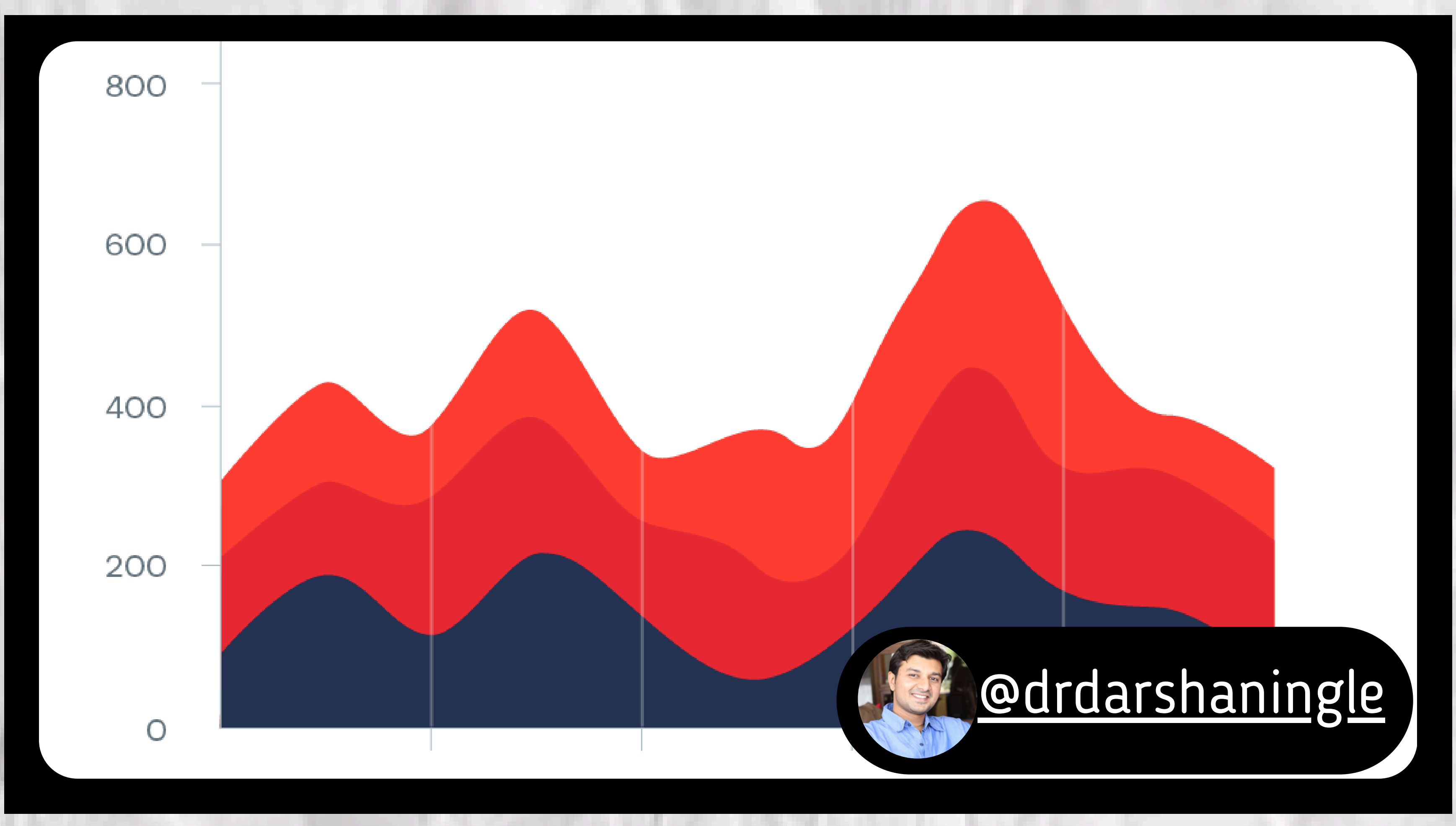
Data Type : Bivariate

USE WHEN: SHOWING THE COMPOSITION OF A WHOLE ACROSS CATEGORIES.

DON'T USE: WHEN LIMITED CATEGORIES, FOCUS IS ON OVERALL TRENDS.

EXAMPLE : SALES BY PRODUCT CATEGORY AND REGION.

AREA CHART



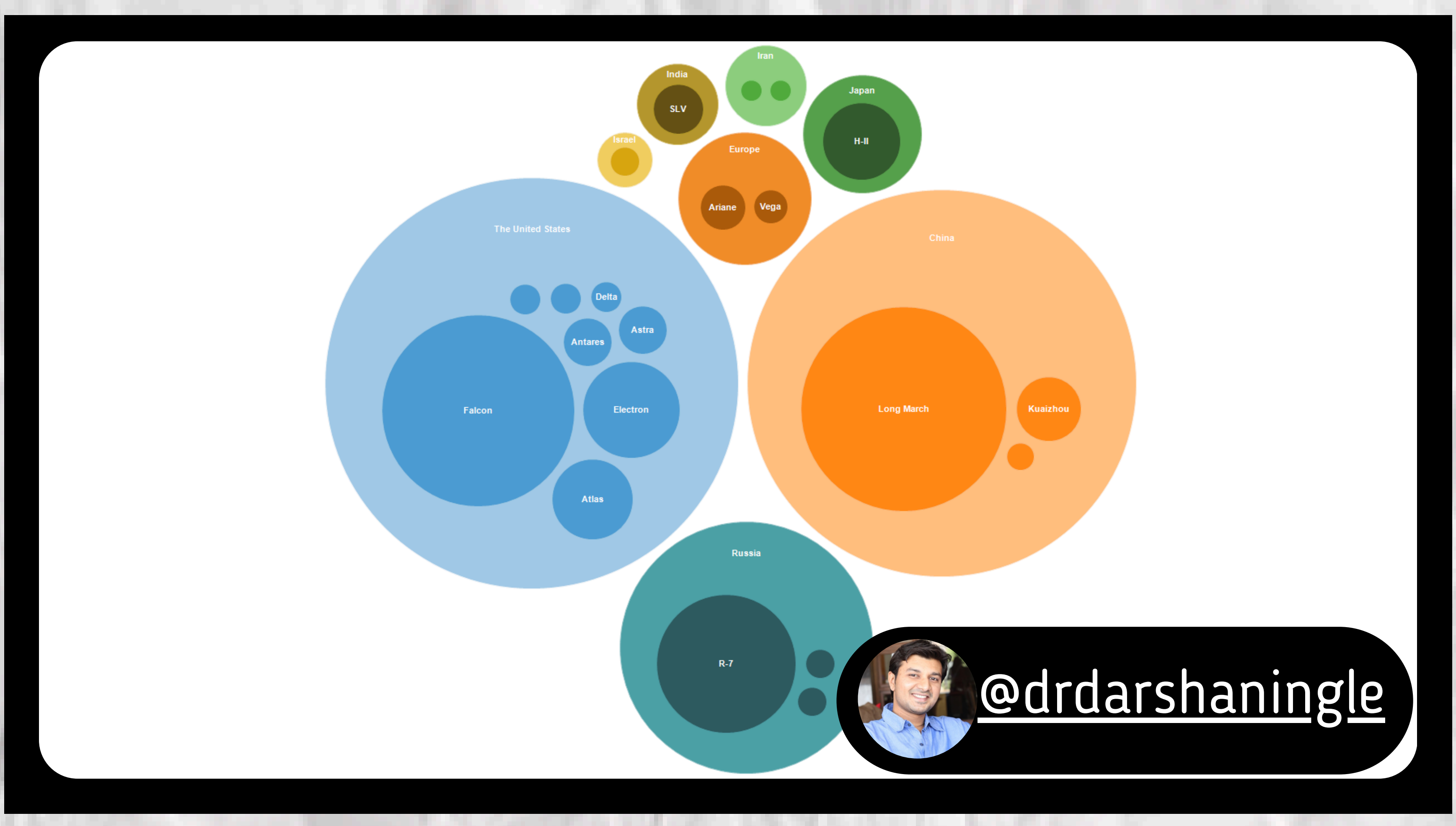
Data Type : Bivariate

USE WHEN: SHOWING TRENDS OVER TIME WITH EMPHASIS ON MAGNITUDE.

DON'T USE WHEN: HIGHLIGHTING SPECIFIC DATA POINTS IS DIFFICULT.

EXAMPLE : STOCK PRICE FLUCTUATIONS OVER TIME, REVENUE GROWTH BY QUARTER.

BUBBLE CHART



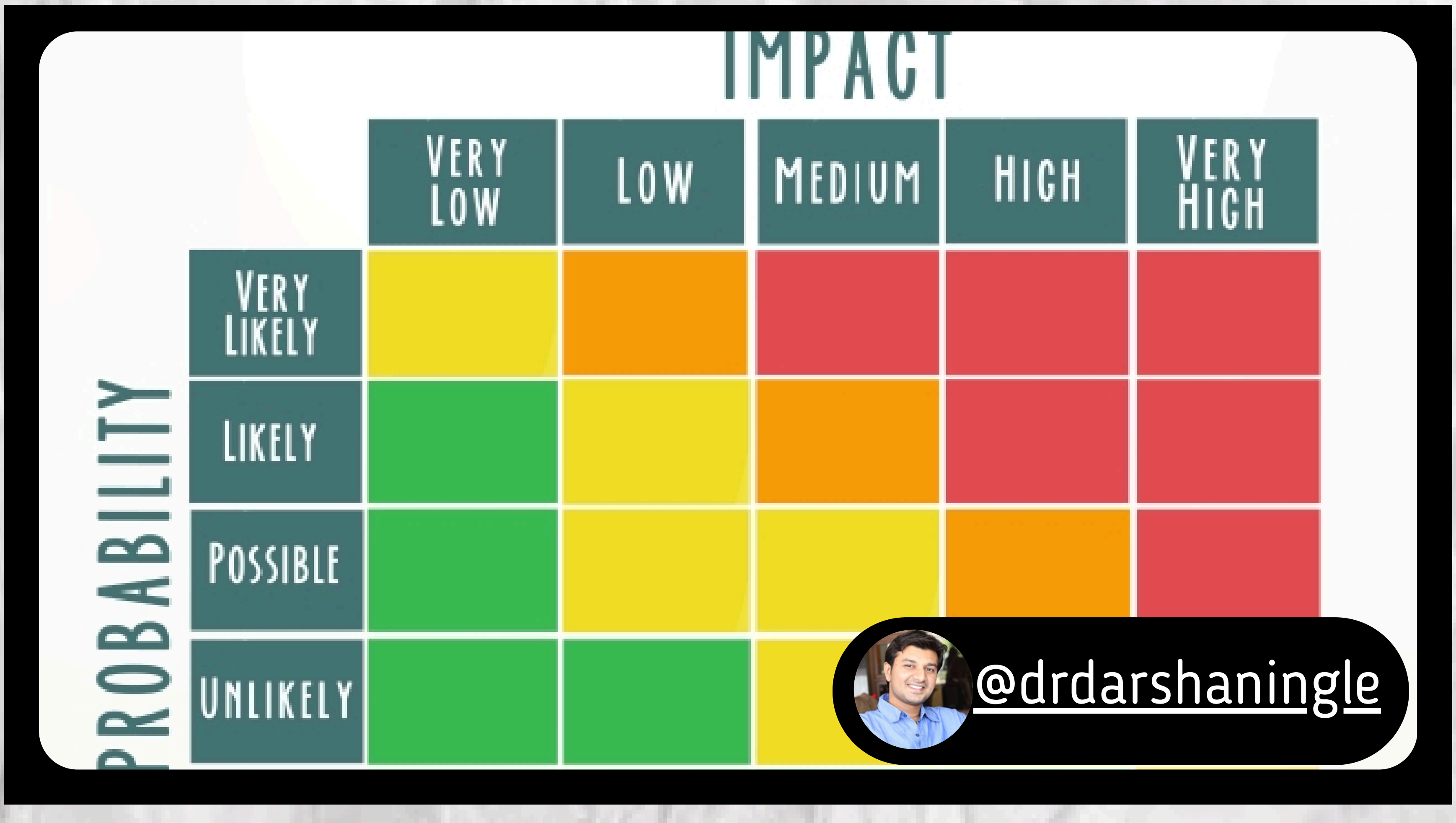
Data Type : Bivariate/Multivariate

USE WHEN: SHOWING RELATIONSHIPS BETWEEN THREE VARIABLES.

DON'T USE WHEN: COMPLEX DATA, MANY BUBBLES CAN CLUTTER THE CHART.

EXAMPLE : SOCIAL MEDIA ENGAGEMENT BY PLATFORM, SENTIMENT ANALYSIS (SIZE INDICATES MESSAGE VOLUME).

HEATMAP



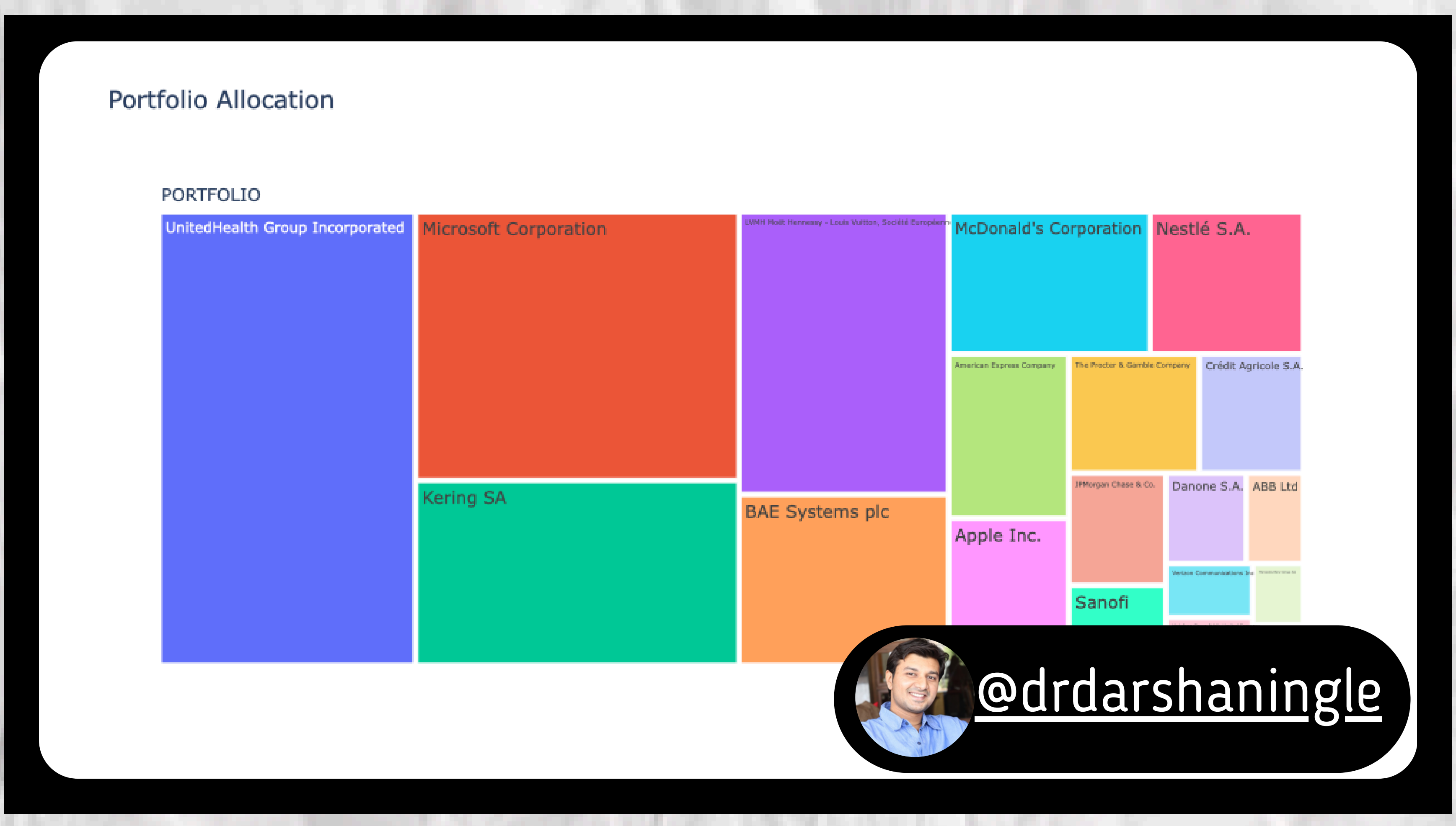
Data Type : Bivariate/Multivariate

USE WHEN: REPRESENTING DATA AS COLOR INTENSITY IN A GRID, USEFUL FOR SHOWING CORRELATIONS ACROSS MANY DATA POINTS.

DON'T USE WHEN: PRECISE COMPARISONS OF INDIVIDUAL VALUES.

EXAMPLE : CUSTOMER SENTIMENT ANALYSIS ACROSS PRODUCTS AND PRICE RANGES, STOCK MARKET CORRELATIONS.

TREEMAP



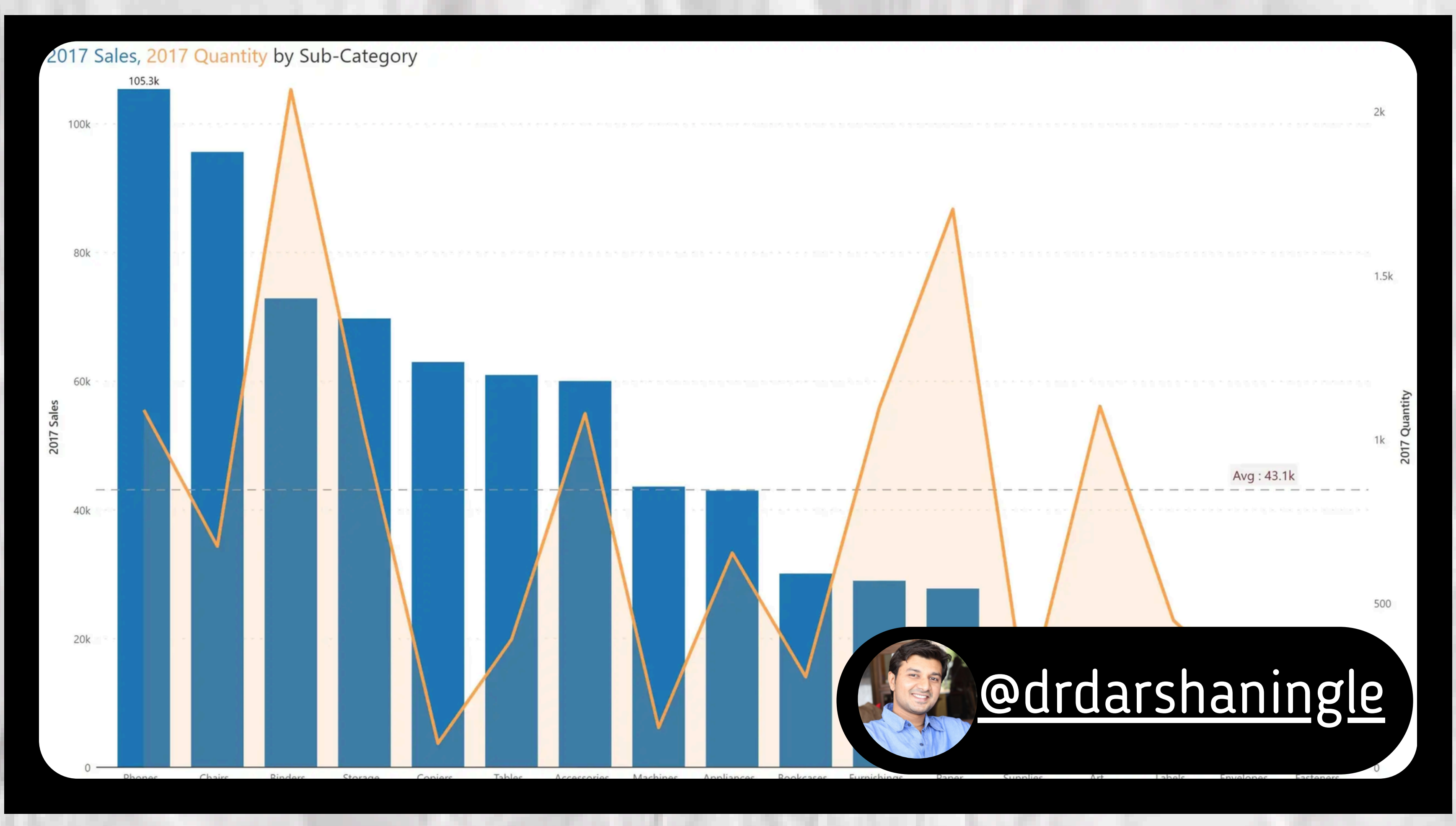
Data Type : Bivariate/Multivariate

USE WHEN: VISUALIZING HIERARCHICAL DATA STRUCTURE AND PROPORTIONS.

DON'T USE WHEN: LIMITED FOR LARGE DATASETS OR MANY LEVELS.

EXAMPLE : COMPANY DEPARTMENT STRUCTURE AND BUDGET ALLOCATION.

DUAL AXIS CHART



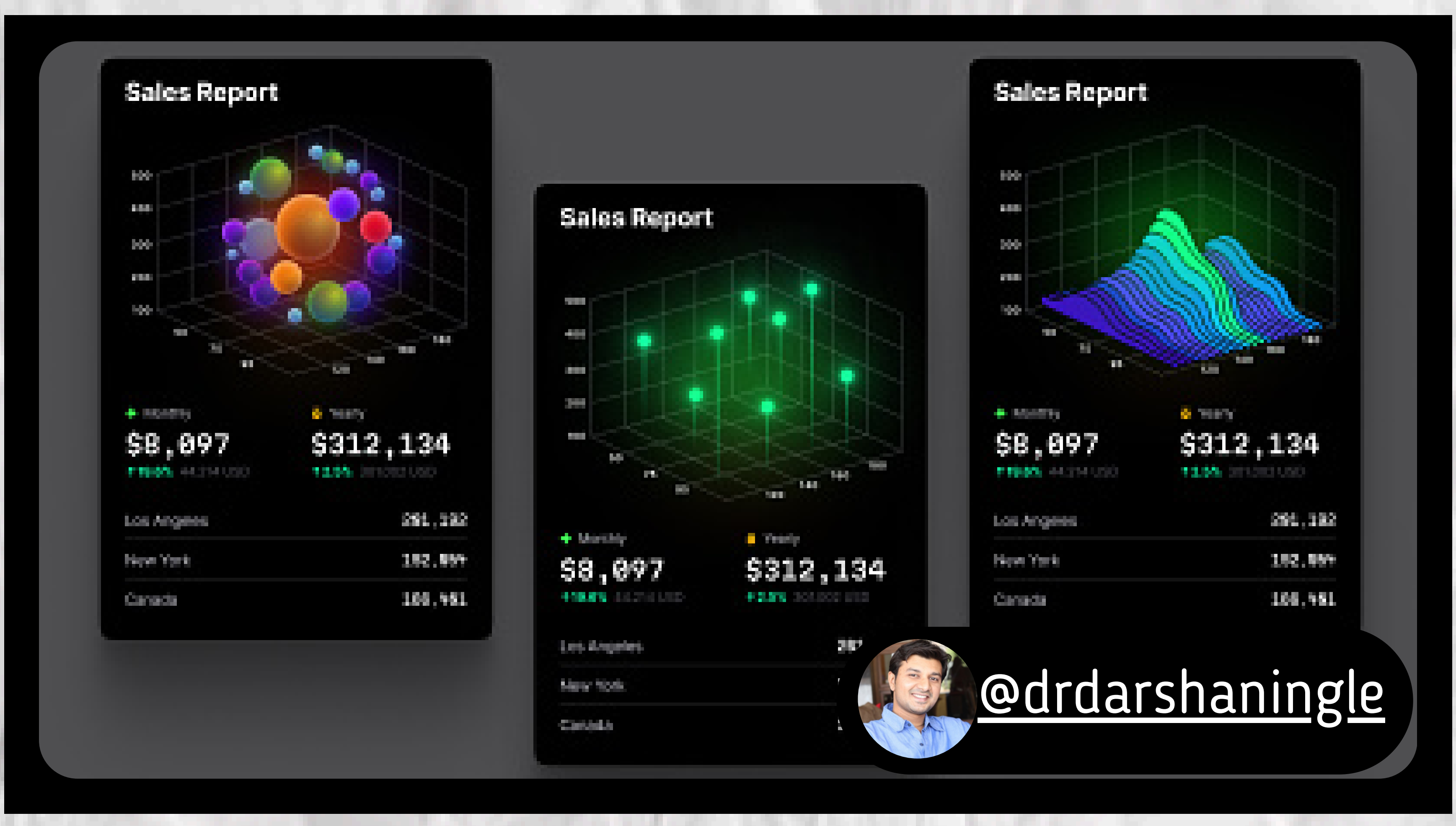
Data Type : Bivariate

USE WHEN: PLOTTING TWO DIFFERENT MEASURES ON THE SAME CHART WITH SEPARATE Y-AXES.

DON'T USE WHEN: COMPLEX TO INTERPRET, AVOID FOR CASUAL AUDIENCES.

EXAMPLE : SALES FIGURES AND CUSTOMER SATISFACTION RATINGS OVER TIME.

3D CHART



Data Type : Bivariate/Multivariate

USE WHEN: CAN ADD VISUAL INTEREST, BUT MAY REDUCE CLARITY.

DON'T USE WHEN: DATA CAN BE DIFFICULT TO PERCEIVE IN 3D, AVOID FOR COMPLEX DATASETS.

EXAMPLE : 3D PIE CHART FOR BUDGET BREAKDOWN (CONSIDER ALTERNATIVE IF UNCLEAR).